

YASHI HUANG

Ph.D. Candidate in Operations Management | AI, Healthcare Analytics & Decision Sciences

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ACADEMIC PROFILE

Ph.D. candidate in Operations Management with research and industry experience at the intersection of artificial intelligence, healthcare operations, business analytics, and operations management. Research combines machine learning, probabilistic forecasting, simulation, and optimization with real healthcare applications at a large U.S. cancer institute.

EDUCATION

- University of Rochester, Simon Business School - Ph.D.** Sep 2021 - Dec 2026 (Expected)
Candidate, Operations Management
• GPA: 3.9/4.0. Research areas: AI/ML, healthcare operations, predictive analytics, simulation, and optimization.
- Columbia University - M.S., Operations Research** Sep 2018 - Feb 2020
• GPA: 3.9/4.0.
- Wuhan University - B.A., Financial Engineering** Sep 2014 - Jun 2018
• GPA: 3.8/4.0.

TEACHING EXPERIENCE

- Lab Instructor, OMG 402 - Operations Management,** Spring 2025
University of Rochester
• Supported student learning in core operations management topics and quantitative decision-making, helping connect analytical methods to applied business problems.
- Lab Instructor, GBA 465 - Introduction to Python** Spring 2024
Programming, University of Rochester
• Led two hands-on lab sections in introductory Python programming, guiding students through coding exercises, debugging, and applications of Python to business analytics.

TEACHING INTERESTS

AI in Healthcare; Machine Learning and AI for Business; Healthcare Operations; Data Management and Analytics for Decision Making; Operations Management; Introduction to Python Programming.

SELECTED RESEARCH

- Dynamic Interval Prediction of Waiting Times in a** Jul 2023 - Present
Cancer Care Facility
• Study dynamic waiting-time prediction in appointment-based service systems, focusing on interval rather than point predictions and developing evaluation metrics that jointly capture reliability and informativeness.
• Compare distributional models, including random survival forests, neural networks, and a recurrent marked temporal point process that learns from the full event history, with point-prediction models.
• Construct calibrated intervals using adaptive quantile selection and conformal prediction.
• Develop an adaptive interval-construction method that yields intervals about 5 minutes narrower than standard bands at matched coverage and a real-time updater that improves accuracy by about 5%.
• Use nested simulation to benchmark attainable performance and explainable AI to verify operationally meaningful prediction drivers.
- Probabilistic Forecasting of Intraday Occupancy in a** Oct 2024 - Present
Large Cancer Center
• Study probabilistic forecasting of infusion-chair occupancy in cancer care, where the input is an arbitrary set of provider sessions and the output is a full intraday occupancy curve, including expected values and quantiles.
• Propose an aggregation-disaggregation framework that combines prediction primitives across organizational levels and time resolutions, and evaluate implementations based on machine learning, time-series, and simulation models using real-world data.
• Collaborate with hospital planners to build a decision-support tool that assesses how changes in provider-session timing affect downstream infusion-chair occupancy and operational capacity.

Predictive Scheduling in Service Systems with Simulation and Optimization

Mar 2023 - Present

- Study whether the realization of an appointment schedule can be predicted before the service day begins, with emphasis on how prediction accuracy should be measured and how much predictability exists at the start of the day.
- Develop a simulation-optimization framework that simulates many realizations of the service day and selects the feasible Gantt chart that best represents those outcomes.
- Benchmark the predicted schedule against both simulated realizations and the original appointment schedule to quantify the value of prediction beyond the planned schedule itself.
- Use controlled experiments to examine how predictive performance changes with key sources of operational uncertainty, including arrival punctuality, service-duration variability, and patient no-shows.

HEALTHCARE & PROFESSIONAL EXPERIENCE

Data Scientist Intern - Dana-Farber Cancer Institute, Boston, MA

May 2025 - Aug 2025

- Took the wait-time prediction model from research prototype to the institute's first production machine-learning model for operations, deployed in clinical workflow.
- Built the data pipeline on Databricks (PySpark, Delta Lake) that integrates multiple sources across 1M+ patient records into operational features such as appointment schedules, real-time queue metrics, and provider statistics.
- Productionized the model with MLflow: automated retraining, drift monitoring, and post-launch metrics for accuracy, fairness, and reliability, with tests and documentation for long-term maintenance.
- Worked with 15+ clinicians, product managers, and engineers to fit predictions into daily workflow, and built dashboards for clinical leadership.

SELECTED PRESENTATIONS

Oct 2025: "Dynamic Interval Prediction for Patient Wait Times in a Cancer-Care Facility," INFORMS Annual Meeting, Atlanta, GA.

Jul 2024: "Dynamic Interval Prediction for Patient Wait Times in a Cancer-Care Facility," INFORMS MSOM Conference, Minneapolis, MN.

Aug 2023: "Physician Rostering Problem with Downstream Capacity Constraints," INFORMS Healthcare Conference, Toronto, Canada.

TECHNICAL EXPERTISE

- Programming & Platforms: Python, SQL, R, MATLAB, VBA; PySpark, Databricks, Snowflake, MLflow, Delta Lake, Gurobi.
- AI & Analytics: regression and classification, neural networks, survival analysis, probabilistic forecasting, time-series models, NLP/text mining, explainability.
- Operations & Decision Sciences: scheduling, predictive analytics, queueing and service systems, mathematical optimization, simulation.